

Building Resilience for an Uncertain Drinking Water Future

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Abstract – Enhancing drinking water resilience has become increasingly important. However, a comprehensive analysis of drinking water emergency countermeasures is lacking. This study evaluated eight countermeasures including monitoring, local alternatives, reclaimed water, interconnection, bulk water, pre-packaged water, emergency treatment, and isolation valves from resilience and sustainability (i.e., life cycle cost) perspectives. While countermeasures such as interconnections perform relatively well from both perspectives, there is a clear tradeoff between resilience and cost. Local alternatives and emergency treatment respond quickly and provide sustained supply during emergencies but may incur higher costs. Bulk water and pre-packaged water are typically inexpensive but have limited supply capacity and take time to distribute. As future threats are likely to become more frequent and prolonged, it is prudent for service providers to invest in countermeasures that perform well in both resilience and cost and use an integrated approach that combines high capital projects with bulk/pre-packaged water contracts.

Article Impact Statement – Eight emergency drinking water countermeasures were reviewed and compared from resilience and life cycle cost perspectives.

Keywords: drinking water emergency planning and response; countermeasures; infrastructure resilience; life cycle cost assessment

1. Introduction

Drinking water systems are considered one of the lifeline infrastructure systems in the United States because of the essential functions they provide (The White House, 2013; USDHS & USEPA, 2015). However, their services can be vulnerable to natural and manmade hazards (ASCE, 2021; Maiolo & Pantusa, 2018) resulting in far-reaching public health, economic, and environmental consequences (Clark, 2014; Maiolo & Pantusa, 2018). In face of the projected increases in the frequency of natural and manmade hazards, uncertainties in climate change, infrastructure aging, and the increasing infrastructure development in high-risk locations, great emphasis has been placed on strengthening the resilience of drinking water systems through effective risk management (Pamidimukkala et al., 2021; USCB, 2021). Infrastructure system or water infrastructure system resilience has been broadly defined as the ability to withstand, adapt to, and rapidly recover the system's function from a potentially disruptive event (AWWA, 2021; NIAC, 2009). More specifically, the U.S. Environmental Protection Agency (EPA) defines the resilience of drinking water systems as "the ability of the human organizations that manage water to design, maintain, and operate water infrastructure (e.g., water sources, treatment plants, storage tanks, and distribution systems) in such a way that limits the effects of disasters on the water infrastructure and the community it serves, and enables rapid return to normal delivery of safe water to customers" (USEPA, 2015b). In an effort to balance considerations for risk and resilience in drinking water systems, §2013 of the U.S. America's Water Infrastructure Act (AWIA) of 2018 (Public Law 115-270) amended §1433 of the Safe Drinking Water Act (SDWA) to require community drinking water systems serving 3,300 or more persons to 1) conduct a risk and resilience assessment and 2) develop or update (review and update as needed) an emergency response plan no less than every 5 years (SDWA §1433, 2018). SDWA §1433 states that utilities may apply the most recent voluntary consensus standards (e.g., American Water Works Association (AWWA) J100, G430, and G440) as a means to facilitate compliance, which often involves application of countermeasures (AWWA, 2020, 2021, 2022). Countermeasure was defined in AWWA J100 as "an action, device, or physical element that reduces risk by affecting an asset, threat, or vulnerability" (AWWA, 2021). Countermeasures may be directed at providing detection, deterrence, devaluation, delay, or response (AWWA, 2021).

Previous studies of drinking water system resilience have primarily focused on countermeasure identification and feasibility analysis (Adepu et al., 2022; Anwarul Abedin et al., 2019; Haxton et al., 2018; Nguyen et al., 2021; Pamidimukkala et al., 2021; Weisman & Epting, 2011), resilience assessment framework development (Ayala-Cabrera et al., 2018; Klise & Murray, 2018; Liu et al., 2021; Morley, 2019; Shuang et al., 2019; Zhan et al., 2020), and comparisons of system and technology design (Nikolopoulos et al., 2021; Schoen et al., 2015). Despite these efforts, a comprehensive review of countermeasures and their influence on drinking water resilience is still lacking. Sustainability is also a key factor to consider when making decisions about planning and selection of countermeasures, which can often be at odds with resilience. While resilience emphasizes redundancy, reliability, response, recovery, and diversity (Kharrazi et al., 2020), sustainability emphasizes efficiency, effectiveness, and resource conservation (di Fabio, 2017). Sustainable countermeasures that support provision of safe and reliable drinking water must consider minimizing economic cost, environmental resource impact, and societal impact (Clune & Zehnder, 2020; USEPA, 2012c). Assessing the sustainability of countermeasures necessitates a thorough examination of economic implications, where life cycle

cost analysis serves as a pivotal metric, offering a comprehensive view of financial impacts over time (USEPA, 2012c).

Previous drinking water resilience studies rarely include sustainability as a dimension of evaluation, despite the noted relationship between sustainability and resilience (Krueger et al., 2020; Liner et al., 2012). However, an urban water supply system is the most desirable when resilience enhancement aligns with sustainability goals, such that resources can be effectively utilized for maintaining the desired level of service in the long run. Previous efforts to quantify the sustainability measures (e.g., carbon footprint, life cycle cost) of individual technologies include countermeasures for source water supply limitations, such as water reclamation and desalination (Arden et al., 2021; Zhou et al., 2014). However, these studies focused on long-term planning and preparation rather than the actual response to emergencies. AWWA J100 and Federal Emergency Management Agency (FEMA) Hazard Mitigation Assistance Guidance provided methodologies to conduct benefit-cost analysis (BCA) of countermeasures (AWWA, 2021; FEMA, 2015). AWWA J100 provides a high-level BCA screening method and FEMA Hazard Mitigation Assistance Guidance focused on community-based BCA on a larger scale. Although these methods consider the tradeoff between resilience and economic cost, they do not directly incorporate a life cycle view when assessing the cost. Several other studies have investigated the resilience and life cycle cost tradeoffs among various water distribution network designs (Assad et al., 2021; Piratla, 2016; Yazdani & Jeffrey, 2011). These studies considered one countermeasure (e.g., water distribution redundancy through robust pipe network design); a comprehensive analysis/comparison of different countermeasures from both resilience and sustainability perspectives was found to be lacking.

The goal of this study is to increase our understanding of the resilience and life cycle cost performances across countermeasures commonly applied in drinking water systems. This paper reviews and compares the resilience benefits and life cycle cost (as a preliminary measure of sustainability given limited literature data on other sustainability measures) of four groups (eight types) of countermeasures. This review is an initial step towards understanding of resilience and sustainability synergies and tradeoffs, which facilitates prioritization of potential countermeasures and provides insight for drinking water managers to best allocate limited resources for capital improvement projects (AWWA, 2021).

2. Natural and Manmade Threats to Drinking Water Systems

Drinking water systems are vulnerable to natural and human-caused emergency events (Clark, 2014; Maiolo & Pantusa, 2018; McNabb & Swenson, 2023). Some of the most impactful nature disasters akin to drinking water systems in the United States include floods, hurricanes, droughts, earthquakes, and wildfires (AWWA, 2018; FEMA, 2023a). The scale and the severity of the impact is often tied to the size of the drinking water system and the social vulnerability of its serving communities (FEMA, 2023b; Routley, 2022). Approximately 99% of U.S. counties are impacted by flooding (FEMA, 2021), and for the past 25 years, urban flooding events have occurred every 2-3 days in the United States (UMD & TAMUG, 2018). Flooding events can increase water contamination risk and interrupt operation of water treatment facilities (USEPA, 2015b, 2022a). High winds and heavy rainfall associated with hurricanes or severe storms can result in power loss, flooding, and physical damage to drinking water infrastructure (AWWA, 2018; USEPA, 2022a). For example, Hurricane Katrina in 2005 affected more than 1,220

drinking water systems across Louisiana, Mississippi, and Alabama, and approximately 70% of these facilities remained inoperable for two weeks following the event (Copeland, 2005). The 2021 Winter Storm Uri left approximately 49% of Texans with water service disruptions due to power loss (UH, 2021). On average, approximately 26% of land area in the United States experiences drought of at least moderate intensity in any given year (Folger, 2017). Droughts that occur during the peak water demand seasons can cause water shortages, low system pressures, boil-water advisories, and a potential need for imported water sources (Clark, 2014; Maiolo & Pantusa, 2018; McNabb & Swenson, 2023). Earthquakes, though major ones with a 7.0 magnitude had a low frequency of occurring once every 60 years, can cause significant physical damage to drinking water infrastructure and disrupt service (AWWA, 2018; USEPA, 2018a). For example, the 1994 Northridge Earthquake in Los Angeles damaged 7 reservoirs, over 60 transmission mains, and 1,000 distribution pipes (Folger & Cody, 2015). Wildfires are one of the most common natural hazards in the United States (FEMA, 2023a), although they typically do not directly impact drinking water systems. When they do, it can result in short-term variations in source water quality (CWN & WRF, 2014), cause physical damage to water infrastructure, and/or increase operational and planning complexity for water providers (AWWA, 2018; Page et al., 2022). For example, in the 2017 Tubbs Fire in Santa Rosa and the 2018 Camp Fire in Butte County, California, the thermal degradation of plastic pipes resulted in the release of volatile organic compounds (e.g., benzene) into the water system, leading local water providers to isolate the affected area and issue a water advisory (Proctor et al., 2020; SRW, 2018).

Likewise, municipalities need to remain vigilant regarding potential human-caused events such as sabotage, terrorism, and unintentional contamination (USEPA, 2021a). Sabotage is an intentional and malicious physical act involving interfering with a utility process and may result in the loss of water treatment and supply capacities (AWWA, 2021; USEPA, 2021a). Water sources and infrastructure may also be impacted by pre-planned terror threats, including chemical, biological, or radiological contamination; physical damage; and cyber incidents (Gleick, 2006; States, 2009). These acts could significantly impair drinking water operations and compromise potable water service. Unintentional contamination can be the result of a multitude of non-malicious events, such as a hazardous chemical containment breach that enters either the drinking water source or distribution network. Since 2010, more than 281,000 chemical release incidents have been reported in the United States (USEPA, 2021b); some of which have led to drinking water contamination for large populations resulting in negative impacts on public health and safety (e.g., the 2014 Elk River chemical spill) (Rosen et al., 2014).

3. Selected Countermeasures and their Contribution to Resilience

In this section, four groups of commonly applied drinking water countermeasures are discussed from a resilience perspective, including online water quality monitoring, alternative drinking water sources, emergency treatment systems, and the use of isolation valves to limit the extent of distribution water contamination. The advantages and disadvantages, the key design considerations, and the emergency response procedure related to each countermeasure are examined. Lastly, the resilience performances of the countermeasures are compared.

3.1. Online Water Quality Monitoring

Online water quality monitoring systems contribute to resilience via early detection, which enables operators to quickly and effectively respond, prevent, and control potential impacts to

drinking water systems (Benedict et al., 2017). An online water monitoring system may use smart sensors, instrumentation, and controls to evaluate real-time water quality and, when specific conditions arise, alert an authority. It is considered the most beneficial for combatting water quality-related emergencies (USEPA, 2015a). For example, after the 2012 Hewlett wildfire in Colorado, Fort Collins Utilities deployed a water quality monitoring system upstream of the Poudre River water supply, providing real-time data on turbidity, conductivity, and temperature (Oropeza et al., 2013). Improvements in sensors and other associated technologies have resulted in online monitoring capabilities that are more accurate, timely, and affordable (Adu-Manu et al., 2017). However, sensors require routine maintenance and/or replacement due to rust or damage associated with long-term exposure to water (Jan et al., 2021).

Key design considerations of online water quality monitoring systems include sensor placement, monitoring parameters, and monitoring station configuration. Sensors should be placed in secure locations that are representative of source water and/or distribution system drinking water (Wagner et al., 2000). Typical monitoring locations include strategic locations in the watershed and source water body, raw water intake points, operational control points, distribution system entry points, interconnections, mixing zone, and critical customer connections (USEPA, 2016, 2018c). Water quality parameters to monitor should be selected based on parameters capable of supporting contaminant detection. For source water, the most commonly monitored parameters include turbidity, oxidation-reduction potential (ORP), temperature, pH, and electrical conductivity (Adu-Manu et al., 2017; Geetha & Gouthami, 2017; Jan et al., 2021; Lambrou et al., 2014; Priya et al., 2017). For distribution systems, the most commonly monitored parameters include chlorine residual, pH, specific conductance, temperature, dissolved organic carbon (DOC)/total organic carbon (TOC), ORP, and spectral absorbance (USEPA, 2005b, 2005a, 2009, 2018c). Monitoring stations typically consist of 1) instrumentation for water quality measurement; 2) computing system for data transfer, processing, and remote control; 3) communication system for instruction and data transmission between the control center and monitoring stations; 4) power supply and distribution; and, 5) station housing and ancillary equipment (USEPA, 2016, 2023b).

Once an alert is triggered by the online water quality monitoring system, the first step is to initiate an alert investigation process to eliminate false alerts caused by equipment malfunctions and data analysis issues. After alert validation and verification, designated personnel are notified to take response actions. Common response actions include: 1) closing intake valves, 2) modifying treatment to remove or neutralize potential contamination, and 3) activating a Contamination Response Plan and Risk Communication Plan (USEPA, 2016, 2018c). Performance of online water quality monitoring systems can be assessed based on mean time to detection, percentage of time the system is fully operational, and data accuracy and data completeness (USEPA, 2014). The CANARY system installed at the Greater Cincinnati Water Works, a well-funded EPA pilot project, for example, takes a median time of 1.6 hours to produce an alert (based on consecutive timesteps of abnormal data) and is fully operational 81.7% of the time. It also does not provide perfect data. On average, data accuracy of the CANARY system was 96.54%, and data completeness was 93.1% (USEPA, 2014).

3.2. Alternative Drinking Water Sources

Alternative drinking water sources contribute to resilience through serving as a backup supply to avoid or minimize interrupted water service to customers (USEPA, 2011). Alternative water sources may include those for long-term water supply resilience, such as local alternative ground and surface water sources (including relocation of water intakes), reclaimed water, and interconnections with neighboring water providers, and those for short-term emergency responses, such as bulk water and pre-packaged water (SDWA §1433, 2018; USEPA, 2011, 2023a). Long-term options are instrumental in guaranteeing a sustained water supply, serving as a robust buffer against unexpected interruptions and persistent challenges like droughts. In contrast, short-term options offer immediate, although temporary, relief by providing alternative water delivery methods during service disruptions. It should be noted that while preservation and conservation of existing water sources (e.g., construction of flood protection barriers (SDWA §1433, 2018)) is also an important countermeasure, it is not specifically discussed in this paper given that limited actions are required by these countermeasures during the response stage. Alternative drinking water sources have been widely used for disaster relief, including storms (FDA, 2022), droughts (H.R.4099, 2021), flooding (CDC, 2021b), as well as in response to water contamination events (Wang et al., 2019).

3.2.1. Local Alternative Ground and Surface Water Sources

Groundwater is an alternative water source that may require limited treatment and exist in close proximity to consumers. Aquifer recharge may be an option for additional water storage (OU, 2016). However, groundwater sources can be difficult to locate and access initially. Potential water quality issues related to groundwater mineral contents (e.g., arsenic, fluorides, and nitrates) and over-pumping must also be considered (OU, 2016). Alternative surface water sources, on the other hand, may have easier initial access, but can require more extensive treatment. Surface water source availability may also vary by season and climate. Developing surface water sources may require construction of dams and reservoirs for storage (ATL, 2022; ICPRB, 2023; Tacoma Water, 2018), which can sometimes be costly and/or have significant environmental impacts.

Key considerations when developing local alternative water sources include independence from the current intake, compatibility with and distance from the current system, and water availability. For use as an emergency backup water source, local alternative water sources need to be independent from the current primary water intake (WA DOH, 2017a) and compatible with the existing water supply system (OU, 2016). The alternative water source also needs to be evaluated to determine the potential amount of current demand that can be met with consideration of seasonal variations (Ngasala et al., 2018) and distance from existing infrastructure in relation to cost (Marsden et al., 2006; OECD, 2007). The volume of water that an alternative source can provide during an emergency can be calculated as the amount of water in storage plus the volume supplied by alternative water supply minus current system demand. It is vital for the volume of alternative water to be sufficient for several days to months depending on the level of resilience deemed adequate for local and regional environmental conditions, potential hazards and associated risks, and feasibility of implementation. The duration of emergency supply may be extended by issuing customer and public water use restrictions such as Do Not Drink orders (CDC, 2022; USEPA, 2004). However, in the event of certain disasters such as droughts, floods, and earthquakes, multiple water sources may be compromised and local alternative sources may not be an option (USEPA, 2011). An operational local alternative water supply can typically be utilized within the hour to provide backup water for a community and is

typically suitable for emergencies lasting longer than 3 days but less than 1 month (USEPA, 2011). Local alternative sources are often subject to prior approval by state agencies to ensure safe emergency water consumption (DOE, 2022; NHDES, 2019; WA DOH, 2017b). In the event contamination is already present in the water system, partial or whole system flushing and/or disinfection may be required before an alternative water source is considered safe for consumption (SDWF, 2017; SWRCB-DDW, 2015). Risk communication, public notification, and issuing water use restrictions may be required for effective use of alternate water source implementation when alternative sources are in operation (USEPA, 2018b).

3.2.2. Reclaimed Water

Water recycling, such as non-potable or indirect potable reuse, may be used to alleviate water shortages in an emergency. For example, to alleviate impacts from a 287-day drought in Fukuoka, Japan, the municipal authorities implemented a water recycling facility to produce reclaimed water for toilet flushing and landscape irrigation using a dual distribution system (Takeuchi & Tanaka, 2020). A more sustainable solution that allows for various levels of water use may require direct potable reuse (DPR), which introduces treated wastewater with potable quality directly into a municipal water supply system (USEPA, 2012b). The use of DPR is extremely limited in the United States primarily due to lack of public support. Examples of existing DPR projects (USEPA, 2012b) include those in the Cities of Big Spring (Cabrera, 2018), El Paso (Browne, 2019), and Carroll, Texas (WaterWorld, 2019). Given the amount of water available through the DPR can be limited, water system operators may need to strategically prioritize the use of water provided through DPR when other water sources become unavailable during an emergency.

Wastewater reuse regulations and practices vary by U.S. state, typically overseen by state primary agencies. Those in more arid areas (such as Arizona, California, New Mexico, and Texas) typically have more specific and comprehensive regulations for wastewater reuse (Martinez, 2019). Some states adopt the International Plumbing Code (IPC) regarding construction, installation, alteration, and repair of non-potable reuse systems (ICC, 2014). These regulations, however, do not currently address the effective utilization of reclaimed water resources during emergencies, which can be a potential area for further research and development. Water providers can also benefit from information regarding the quantity of reclaimed water available locally, potential users, and public acceptance. Non-potable reclaimed water is typically delivered through a dual piping system to prevent cross-contamination with potable water. Reclaimed water pipes are typically purple in color and labeled as “NONPOTABLE WATER, DO NOT DRINK.” (WBDG, 2022; WMWD, 2011). Non-potable reclaimed water can also be delivered by water trucks for construction (dust suppression or soil compaction), street cleaning, and other purposes. Truck drivers and reclaimed water users are subject to mandatory training regarding proper and safe handling of reclaimed water. In addition, reclaimed water signage is typically affixed to the water truck (SD PUWW, 2015; CA Water Boards, 2020).

3.2.3. Interconnection with Neighboring Water Providers

Interconnection with neighboring water providers allows water providers to share water resources during an emergency (USEPA, 2023a). Interconnections between water providers allow for more effective coordination and management of water resources by reducing the risk of

over-abstracting local water resources (ICE, 2015). Interconnections are limited in their ability to address broad-scale emergencies that affect multiple regions, such as droughts and earthquakes (Parliament, 2018), and their planning and implementation may take multiple years. Water providers also need to consider potential elevation changes among interconnecting piping networks and potential water loss during transfer (Earth & Water Law, 2017; Parliament, 2018).

Interconnections typically require pre-planning and written agreements between involved parties (USEPA, 2011). Interconnections may be utilized quickly (e.g., within an hour) to provide backup water from one water system to another and are suitable for emergencies lasting longer than 3 days to less than 1 month (USEPA, 2011). Considerations include conveyance capacity, surplus water quantity, emergency water supply capacity, water supply purchase costs, ownership and maintenance of the conveyance infrastructure, and conveyance feasibility over time (CADWR, 2016; Zeff et al., 2014). The quantity of surplus water may be determined based on existing water source supply capacities and projected demands of both water system providers (NJDEP, 2007). Selection of interconnection points should consider natural topographical features and presence of existing development and infrastructure such as highways, railroads, and residential housing between the connecting systems (GEFA, 2011). A pressure-reducing valve may need to be installed if the hydraulic grade (HG) difference between the two systems is elevated. Conversely, a booster pump may be needed to provide interconnected water supply (Riley et al., 2007). Water quality is also a key consideration. Differences in water quality between two systems may cause issues in the distribution network, such as loss of disinfectant residual, discoloration, changes in taste and odor, and pipe corrosion, leading to poor water quality at customer taps (GEFA, 2011; Riley et al., 2007). Such differences often manifest from mixing groundwater with surface water, chloraminated with chlorinated water (TECQ, 2023; USEPA, 2023a), water with large pH differences, and fluorinated and non-fluorinated water. These differences may be corrected with water quality adjustments (Riley et al., 2007). Changing hydraulics can also result in mobilizing solids within the pipe network resulting in water quality issues. Hydraulic modeling can assist with evaluating the impact of combining different water sources over time (NJDEP, 2007; Riley et al., 2007). In an emergency, the affected water provider should initiate communication with the connected water provider to activate an emergency water supply connection.

3.2.4. Bulk Water

Emergency bulk water supply typically transfers treated water to designated locations using potable tanker trucks or in water bladders or bags (USEPA, 2011). Supply water may be placed directly into the distribution network, stationed at strategic and convenient locations so customers can fill water containers, and/or be distributed to consumer houses (Maine CDC, 2014; MassDEP, 2020; NHDES, 2020; USEPA, 2011). Bulk water supply is a temporary solution during emergencies that does not require extensive infrastructure or equipment (USEPA, 2011). Water may be susceptible to contamination from use of inappropriate convoys, transfer valves and pipes, and general lack of control of hygiene and sanitation at filling stations/connections or transfer pipes (FL DOH, 2019; WHO, 2011).

Water providers should develop contractual agreements with water haulers in advance to be adequately prepared for emergencies (USEPA, 2011). The contract should include water quantity, quality, and sanitation requirements in addition to roles/responsibilities of each party

(WHO, 2011). Associated equipment must be food-grade to minimize contamination (FL DOH, 2019); designated for potable water use only (Maine CDC, 2014); approved by the National Sanitation Foundation (NSF) (CT DPH, 2008; FL DOH, 2019; MassDEP, 2020), U.S. Food and Drug Administration (FDA) (FL DOH, 2019), and/or U.S. Department of Agriculture (USDA) (FL DOH, 2019); and/or aligns with AWWA standards (FL DOH, 2019; MassDEP, 2020). Transportation and storage tanks should either be completely sealed or have their vents protected to prevent contamination from foreign matter during filling or emptying. Tank hoses, pipes, pumps, and other handling equipment should be visually inspected to ensure that no rust or sediment is present. Some U.S. states require public drinking water systems to obtain bulk water supply from state-approved vendors (Maine CDC, 2014; MassDEP, 2020; NHDES, 2020) and/or in the presence of a certified operator during bulk water transfer if the water is to be introduced directly into a distribution network (NHDES, 2020). In an emergency, the affected water provider should initiate contact with the bulk water vendor to obtain bulk water supply. Bulk water supply is typically used for less than 30 days (Maine CDC, 2014) and can be initiated within hours of an emergency event (Water Direct, 2021).

3.2.5. Pre-Packaged Water

Pre-packaged water such as bottled water can be transported and stored in the affected areas during an emergency (USEPA, 2011). Pre-packaged water can be distributed quickly and is suited as a first-phase response measure before other water supply options are available. However, it might not be sustainable in case of more extended outages due to potentially prohibitive costs (Pentair, 2021; USEPA, 2011).

Pre-packaged water used in emergencies should be obtained from approved and permitted bottlers and distributors. Many U.S. states maintain a list of permitted pre-packaged water companies and suppliers that can be utilized in an emergency (Cal OES, 2014; USEPA, 2011; WVDHHR, 2021). Pre-packaged water contracts should be developed in advance to reduce time and expense associated with acquiring supplies in an emergency. California, for example, maintains a statewide bottled-water contract that specifies price, response time, minimum delivery quantity in an emergency, and responsibilities for ordering entities and bottled water suppliers (Cal OES, 2014). Based on the state of California's contract terms, requests for "emergency" delivery within the vendor's supply capacity are fulfilled within 5 days after order receipt and requests for "urgent" delivery are fulfilled within 12 hours after order receipt (Cal OES, 2014). Logistical considerations include acquisition of equipment necessary to load pre-packaged water pallets on and off trucks or other transportation vehicles, safe storage to prevent contamination, and effective distribution strategies (Cal OES, 2014). Sites utilized for distributing pre-packaged water to the public should have plenty of open space, easy road access, and good lighting (USEPA, 2011). Affected water providers should initiate contact with a pre-packaged water supplier in an emergency event. Purchasing entities should provide detailed shipping addresses of appropriate distribution sites (Cal OES, 2014). For example, during the 2021 Winter Storm Uri, pre-packaged water distribution in the city of Austin, Texas was delayed as a result of receiving locations lacking adequate equipment to receive and distribute palletized water (ATX & TC, 2021). As a result of the 2014 Elk River chemical spill, some households were without an alternative drinking water source for a day or more due to the difficulty of locating alternate water distribution sites and/or the lack of adequate water supplies at distribution sites (CDC, 2014). Pre-packaged water is suggested for use over short time-periods;

however, during the Flint, Michigan Water Crisis, free bottled water was distributed to customers for 816 days (CNN, 2022). An unplanned impact associated with the large-scale use of pre-packaged water in Flint was the large quantity of waste generated that overwhelmed Flint's initial waste management capacity (Wang et al., 2019).

3.3. Emergency Treatment Systems

Emergency treatment systems contribute to resilience as a temporary or enhanced water supply when the original treatment system is compromised/inadequate. Emergency water treatment systems can be adjustments of existing treatment trains and/or separate mobile treatment systems. They can be applied either in the centralized drinking water systems or as point-of-entry or point-of-use systems in consumer houses (USEPA, 2011). Adjustment of existing treatment trains refers to alterations of existing treatment processes for enhanced contaminant removal without substantial infrastructure changes. Mobile treatment systems are all-in-one self-contained portable treatment packages that can be delivered to a site, set up, and taken down easily and quickly. These systems often consist of multiple filtration/adsorption steps and disinfection (USEPA, 2019). Emergency treatment systems have been commonly used in earthquakes (Sturdivan, 2011), floods (Findell, 2018), wildfires (Becker et al., 2018), hurricanes, and chemical spills (Rosen et al., 2014). Critical determinants in selection of emergency treatment systems include quality of untreated water and intended use of treated water (USEPA, 2019). Some of the decision criteria regarding the selection of emergency treatment systems include but are not limited to cost, ease of deployment, ease of use, maintenance, performance, potential acceptance, energy requirements, supply chain requirements, throughput, and environmental impact (Loo et al., 2012). Water providers are recommended to pre-purchase and/or plan for backup treatment systems to avoid potential delays from unfamiliarity with operating equipment (USEPA, 2011).

3.3.1. Adjustment to Existing Treatment Train

Adjustments to existing treatment trains in an emergency event may include those that can be deployed immediately, such as more frequent backwash of rapid filters, use of powdered activated carbon (PAC), increased coagulant dosage, increased oxidant dosage, as well as those that require more time for operational adjustments, such as granular activated carbon (GAC) filters and addition of new oxidant type.

More frequent backwash of rapid filters is commonly the first measure to be taken in response to turbidity increases, typically as a result of flooding or hurricanes, to assist with faster filter bed restoration. It is typically viable if the required backwashing time and frequency would allow for meeting water demand with the desired water quality (Letterman, 1987). However, in the 2018 Austin, Texas flooding event, the Ullrich treatment plant continued to operate at a high backwash frequency in conjunction with a 4-day boiling water order (Findell, 2018). A side problem beyond the water quality issue in this event is the proper discharge of the backwash. Without a line to the wastewater treatment plant, the large amount of backwash generated was backed up, filled storage basins, flooded parts of the plant, and forced the facility to shut down multiple times, which eventually had to be dumped into a local river (Findell, 2018).

Powdered activated carbon (PAC) is commonly used in response to chemical spills (Rosen et al., 2014), wildfires, and algae blooms (Carollo, 2023; Kaczala, 2019) to control problems with

organics and taste and odor constituents, but it is not effective for microbial removal (Loo et al., 2012). PAC can be metered into the water in dry or slurry forms. The dry form is typically better for water providers that use PAC infrequently. PAC is relatively easy to handle and use, quickly deployable (USEPA, 2012a), and adaptable to many types of treatment trains (Crini & Lichtfouse, 2019). Nevertheless, PAC particles can cake filters and result in reduced filter runs (ENVIS, 2022; Loo et al., 2012), and PAC particles may pass through the filter and decolorate the water (PA DEP, 2020). PAC dust in the air can be potentially explosive (ENVIS, 2022), and PAC addition can produce large quantities of sludge and has a high operating cost if used continuously (ENVIS, 2022). PAC was successfully employed to neutralize cyanotoxin in both the 2014 Toledo, Ohio algae bloom event (Kaczala, 2019) and the 2018 Salem, Oregon algae bloom event (Carollo, 2023). In the 2014 Elk River chemical spill, however, PAC combined with increased oxidation dosage was used for less than 4 hours before it was deemed insufficient for treating the contaminated water (Whelton et al., 2015).

Increased coagulant dosage is used primarily for enhanced turbidity reduction and organic matter removal, typically as a result of wildfires (Becker et al., 2018), hurricanes, and floods. For example, during the 2017 Hurricane Harvey, a Texas city effectively treated high levels of hydrophobic natural organic matter by increasing coagulant dosage (Landsman et al., 2019). However, in the 2011 Tennessee winter storm, the Decatur Water Treatment Plant failed to control turbidity despite more than doubling the Polyaluminium chloride dosage and was forced to shut down for 3 days (Qiu et al., 2019). Coagulants can also be added as an additional treatment step for plants that do not have an existing coagulation process, such as through modular treatment systems assembled on-site or mobile units installed on trailers (Dorea, 2009). Coagulant aids, such as lime and bentonite, can be added in emergencies to improve coagulation efficiency (The Water Treatments, 2010). In emergencies, sludge generated is commonly disposed of into waterways, ideally downstream of a treatment plant intake (Dorea, 2009). However, this practice may be undesirable due to high coagulant concentration in sludge which can be toxic to aquatic ecosystems (Arvia Technology, 2022; Dorea, 2009).

Increased oxidant dosage can be used in microbial contamination incidents (Creamer, 2014; Reiss, 2014) caused by floods and hurricanes for enhanced contaminant removal and taste and odor control (Mohammed et al., 2020; Nicholas, 2019). Typical oxidants used in drinking water treatment are chlorine (Cl_2), ozone (O_3), hydrogen peroxide (H_2O_2), chlorine dioxide (ClO_2), and potassium permanganate (KMnO_3) (Crittenden et al., 2012). Oxidation may increase the quantity of oxidation byproducts such as bromate and halogenated organic chemicals (AWWA, 2011). Over-dosage can degrade performance of biological purification processes (EMIS, 2010). It also has a relatively high operating cost (AWWA, 2011) and can cause potential taste concerns when water is treated with chlorine-based oxidants (Loo et al., 2012). In the 2014 E. Coli contamination incident in Wellesley, Massachusetts, the Wellesley water treatment plant flushed the distribution system with chlorine-enriched water, isolated the contaminated water storage tank, and issued a 3-day boil water order (Creamer, 2014; Reiss, 2014). However, in the Flint Water Crisis, the Flint Utilities Department increased the chlorine dosage for bacterial control, which resulted in corrosion concerns and an increased level of total trihalomethanes (CNN, 2022).

Granular activated carbon (GAC) can be used as a filter medium either as a layer in a rapid-sand filter or in a separate filter contactor (ENVIS, 2022) in response to contamination, such as chemical spills (Navy Pier, 2020) and PFAS pollution (Cape Fear Public Utility Authority, 2018; Metcalf, 2021; Natick Town, 2023; NYDEC, 2020; WPW, 2022). GAC is easily handled and used. The use of fixed GAC beds permits easier process control than PAC. However, GAC is approximately two to three times the initial cost of PAC (ENVIS, 2022) and frequent reactivation may be necessary (Brandt et al., 2017). Effluent turbidity, taste and odor, microorganism growth, and contaminant concentration must be monitored to determine whether GAC is operating properly. Head loss must also be monitored to determine when the unit needs to be backwashed (ENVIS, 2022). The use of hybrid zeolite-GAC filtration to treat 5 MGD of contaminated water was implemented in the 2021 Red Hill Fuel Leak in Hawaii (Navy et al., 2022). In response to PFOA contamination, the Village of Hoosick Falls, New York's municipal water system installed a temporary GAC system and operated it for 1 year (HF, 2016).

Addition of a new oxidation type assists with further destruction of contaminants in chemical spills (APTwater, 2023). Common combinations of oxidants include (1) O₃ and H₂O₂, (2) UV and O₃, (3) UV and H₂O₂, and (4) UV and TiO₂ (Crittenden et al., 2012). Technology-specific considerations when introducing a new oxidation type may include problems associated with the combination of O₃ and H₂O₂: (1) volatile organics will be stripped from the ozone contactor; (2) production of O₃ can be an expensive and inefficient process; (3) gaseous O₃ present in off-gas of the ozone contactor must be removed; (4) maintaining and determining proper O₃/H₂O₂ dosages may be difficult; and (5) low pH is detrimental to the process (Crittenden et al., 2012; EMIS, 2010; USEPA, 2006). In response to the methyl tertiary butyl ether (MTBE) contamination in South Lake Tahoe, California, advanced oxidation treatment systems that mix H₂O₂ and O₃ were employed at two drinking water supply wells in 2002 and 2004, respectively, which effectively reduced the contaminant to undetectable levels (APTwater, 2023).

3.3.2. Mobile Treatment Systems

Mobile treatment systems can be deployed and run independently of an existing treatment facility. These systems can operate on-site which reduce issues associated with water transportation. For instance, during the 2010 Baja California earthquake, mobile units were brought onsite by Calxico Water Treatment Plant to treat and bypass the damaged clarifiers and then piped into the finished water reservoir before being distributed (Sturdivan, 2011). Mobile treatment systems combine several treatment technologies and can assist with complex water quality issues. However, mobile treatment systems can be complicated to operate and maintain, and costly to use (USEPA, 2019). Owners of mobile treatment systems include the U.S. military, State National Guard, and/or private sector vendors. For example, the Department of Defense (DOD) and some state National Guard units maintain water purification systems that are typically used for troop support during overseas deployment; these systems can typically be deployed in a domestic emergency in certain circumstances. In addition, the private sector has a wide range of products and experience with mobile treatment systems in disaster scenarios. For instance, a Water-on-Wheels Mobile Water Treatment System (WOW Cart) manufactured by WaterStep can treat up to 14,400 gallons per day (USEPA, 2022b), a Mini Water Treatment Units (Minis) manufactured by WaterStep can treat up to 10,000 gallons per day (USEPA, 2018d), and the PALL Water's Aria FAST mobile treatment system applied in the Calxico earthquake emergency (Sturdivan, 2011) and the Cisco, Texas flooding event (Pall Water, 2018)

can treat up to 1 million gallons per day. Water providers should consider establishing partnerships with local military or reserve units that have mobile water purification systems. A treatment deployment site should have ancillary facilities for water abstraction, power, plumbing, residuals management, and security (USEPA, 2019). In the 2010 Baja earthquake, it took 3 weeks for two mobile water treatment units to be shipped from New York state to Southern California and be hooked up, permitted, and have water flowing, and 3 more weeks for another two mobile treatment units to be manufactured (Sturdivan, 2011).

3.4. Water Isolation Valves

Water isolation valves contribute to water distribution system resilience via stopping or preventing the further spread of water contamination by shutting down distribution zones until problems are rectified. Water distribution isolation valves are typically the most suitable for preventing or reducing the impact of contamination-related emergencies such as chemical spills. The more valves present in a distribution system, the more likely contamination impact can be restricted to a smaller portion of the system with adequate detection lead time (Hwang & Lansey, 2021; Sukru & Mays, 2004). The contaminant needs to be promptly flushed out of the system to minimize absorption into the pipe wall and other appurtenances. A common rule-of-thumb for isolation valve numbers is a minimum of $n - 1$ valves around a junction with n links (pipes) connections (Sukru & Mays, 2004). However, contaminant containment can be complex and challenging given that as each valve closes, the hydraulics of the system change. In addition, back siphonage conditions might occur at customer premises (Selecky et al., 2004). Facilities can either operate manual valve shutdown (operating in-person) or apply remotely controlled isolation valves. Operation of remotely controlled isolation valves requires a sensing element, logic solver, and output signal to close the valve. Manual activation systems can avoid spurious shutdowns that may occur with automatic activation systems; however, automatic activation enables faster water isolation and reduces human error.

In an emergency, water suppliers should take quick actions to isolate portions of the system that are suspected of contamination by closing isolating valves. Since emergencies may not be predicted, water suppliers may not be able to provide advanced customer notification. Affected customers in the isolated area should be notified as soon as possible, including customers outside of the affected area (USEPA, 2013). When shutting off water supply, water hammer issues may occur. Operators should try to open or close a valve slowly to prevent water hammer issues (Tameson, 2021). A remotely controlled smart water valve can be activated within 100 milliseconds; however, the response time can be subject to delays in larger water systems (Antzoulatos et al., 2020). When bringing the water system area back to full service, water pipes in the isolated area should be flushed for a duration determined by parameters such as contaminant, pipe material, pipe diameters, and contact time to ensure water quality and safety (Hauptert & Magnuson, 2021).

3.5. Resilience Comparison of the Countermeasures

While the above countermeasures provide important functions in enhancing drinking water system resilience, each measure has unique merits and drawbacks related to emergency response. The effective implementation of alternative drinking water sources, emergency treatment systems, or water isolation valves for impact reduction depends on successful early detection of potential contamination. Online water quality monitoring provides a means of early detection;

however, it might not be entirely reliable and is vulnerable to potential disruption during power or internet outages. Water isolation valves are the most effective when contamination is still isolable or controllable. The earlier suspected contamination is contained, the more likely the drinking water system can maintain partial or full water service to customers outside of the impacted zone. Once suspected contamination enters the entire distribution network, a system shutdown may be required.

A reliable online water quality monitoring system can enable early detection and actions; successful operation of water isolation valves requires trained personnel to identify the number and locations of the valves to be shut down quickly and accurately. Water isolation valves are limited in addressing water shortage problems in the event of source water contamination. When installed ahead of time, local alternative ground and surface water sources, reclaimed water, interconnections, and emergency treatment systems can be quickly brought online in an emergency event, typically within 60 minutes (USEPA, 2014). In general, alternate water sources can be used as a direct substitute for a conventional supply with the exception of reclaimed water which may be restricted to non-potable use. The duration of an alternate water supply may also be limited, ranging from less than a few hours (USEPA, 2011) to several weeks (Mlive, 2018). In extreme cases, a countermeasure may be implemented for over a year while drinking water infrastructure is upgraded to combat contamination (HF, 2016). Bulk and pre-packaged water typically takes hours to days for effective delivery to consumers (Cal OES, 2014; Water Direct, 2021). While bulk and pre-packaged water quality allows for multiple uses, they are typically used for drinking and cooking given the high cost (NHDES, 2021).

Should short-term contamination of the water supply with a known contaminant occur, emergency water treatment may be possible; however, if contamination from an unknown chemical occurs, emergency water treatment may not be effective due to a lack of information. In this situation, isolation valves and alternative water sources may be the most appropriate early emergency response phase solutions. Water service would be temporarily constrained to prevent further spread of contaminated water and to facilitate necessary flushing of the distribution system, while maintaining pressure for fire suppression. A directive is typically issued to the public to boil water, not to drink tap water, or not to use tap water. An unaffected alternative water source would then be delivered in response to water supply disruption. After screening and identification of the unknown contaminant, emergency water treatment could be brought online.

4. Life Cycle Cost Comparison of the Countermeasures

The life cycle cost of each countermeasure is calculated as a sum of capital, operation and maintenance (O&M), and response cost in the form of equivalent annual cost (EAC, Equation 1) and normalized equivalent annual cost by water supply capacity (EAC_Q, Equation 2). Capital cost is the one-time fixed expenditure associated with acquisition and installation of the countermeasure. O&M cost is the annual, ready-to-serve fee that makes the countermeasure available when emergency situations occur. Response cost is the cost associated with using the countermeasure after an emergency happens. For purpose of this study, the emergency is assumed to be a 14-day event that occurs every 10 years (CDC, 2021a; Cramton et al., 2023; FEMA, 2004). Only costs incurred by the drinking water provider are considered in this study. EAC is the annualized cost of the countermeasure regardless of scale. EAC_Q further normalizes the EAC based on the scale of drinking water supply involved to allow comparability. Cost data

were obtained or estimated based on reports from governments and water providers, journal and news articles, and relevant websites. Detailed calculation data can be found in Table S1 of the supporting information. In particular, O&M cost of local alternatives, reclaimed water, interconnection, and valves are assumed to be zero given these countermeasures are typically used as a part of normal operations beyond the time of emergencies. O&M cost during the manufacturing, storage, and retail phases of pre-packaged water and emergency treatment water systems are assumed to be already included in their prices. Costs related to the distribution of the pre-packaged water are not considered as the distribution may be carried out by entities other than the water providers (e.g. federal, state, and local emergency response agencies) (ATX & TC, 2021). Response cost of local alternative water sources, reclaimed water, interconnections, and isolation valves are assumed to be 2% of the capital cost (OECD, 2005). A cost range is reported when multiple data sources were identified for a specific component. Costs are adjusted to 2021 dollars using the change in the Consumer Price Index between 2021 and the year the cost is realized (USBLS, 2022).

$$EAC = C_k \times \frac{i}{1-(1+i)^{-n}} + C_{OM} + C_R \times d \times p \quad (\text{Equation 1})$$

$$EAC_Q = \frac{C_k \times \frac{i}{1-(1+i)^{-n}}}{Q} + \frac{C_{OM}}{Q} + \frac{C_R \times d \times p}{Q} \quad (\text{Equation 2})$$

where:

EAC = Equivalent annual cost, 2021\$/year;

EAC_Q = Normalized equivalent annual cost, 2021\$/MG;

C_k = Capital cost, 2021\$;

C_{OM} = Annual O&M cost, 2021\$/year;

C_R = Response cost of running the countermeasure for a day, calculated from the annual cost, 2021\$/year;

Q = Water supply capacity of associated drinking water treatment facility, MG/year;

i = Annual discount rate, 2.5 % (USBR, 2020);

n = Expected life span, see the SI for detailed information, years;

p = Assumed probability of an emergency in a given year that triggers response action using countermeasure, assumed to be 10% per year (Cramton et al., 2023); and,

d = Assumed duration of drinking water emergency, assumed to be 14 days (CDC, 2021a; FEMA, 2004).

Figure 1 provides a comparison of EAC and EAC_Q of studied countermeasures. It should be noted that these comparisons are intended to provide a broader view related to the countermeasures' life cycle economic performances based on a set of confined data. Individual water systems will have unique needs, opportunities, and affordability associated with the effectiveness and employment of countermeasures and emergency response capabilities (Sullivan & Schmelling, 2012). Oftentimes, local conditions, including threat expectations, consequence reduction, dual-use benefits (e.g., enhancements to utility operations or community preparedness), resources, and capabilities will govern implementation (Sullivan & Schmelling, 2012). Based on data collected, bulk water and pre-packaged water have the lowest averaged EAC values, followed by emergency water treatment, interconnections, and online water quality monitoring; local water supply alternatives, reclaimed water, and isolation valves have the

highest costs. EAC_Q follows a similar trend with the exception of installing isolation valves associated with a lower cost ranking and emergency water treatment with a relatively higher cost ranking compared to corresponding EACs. This difference is due to cost data for valves were generally sourced from large water treatment facilities and data for emergency water treatment were sourced from small facilities indicating potential viability and use among different size water treatment facilities. The largest EAC and EAC_Q contributor is the capital cost, with the exception that pre-packaged and bulk water's largest cost contributor is emergency response.

Pre-packaged water is more costly than bulk water due to additional cost associated with production. Although bulk water and pre-packaged water costs in past emergency events have typically been considered low, these methods typically do not provide a full level of service to consumers. To meet the entirety of consumer demand, associated costs would be significantly higher. While the flexibility these methods provide and the lower capital investment are highly desirable, it should be noted that the distribution of pre-packaged water may require significant logistical support resources to run Point-of-Distribution operations.

Online water quality monitoring systems are also relatively cost-effective since they do not require extensive construction investment beyond installation of sensors and monitoring equipment. Trained personnel are necessary to operate these systems and interpret data for informed decision-making which may result in additional costs. In addition, false alarms may trigger unnecessary emergency response, which also may incur additional cost and reduce trust in the alarms' reliability. Online monitoring systems may help the drinking water systems to better understand the distribution system hydraulics, improve process control, and enhance customer service; however, they also increase system sophistication that places additional capability demands on water systems and exposes them to additional challenges (Sullivan & Schmelling, 2012). Furthermore, monitoring and surveillance components that are not supported with well-defined response actions will provide little value in mitigating consequences (Sullivan & Schmelling, 2012).

Interconnections with neighboring water providers are typically less expensive than developing local alternative water sources since interconnections typically require less construction investment. However, collaboration between water providers may be subject to regulatory barriers such as obtaining permits from local, state, or federal agencies which can incur staff labor time cost.

The cost of installing isolation valves is relatively high due to associated pipeline alterations. Developing local alternative water sources is typically more costly than installing interconnections. However, alternate water sources allow for control over management and use of water sources without potential inter-provider competition. Emergency water treatment is also relatively costly but provided quick responses to on-site emergency situations and eliminated the need to transport water; however, these systems may not be effective for all contaminants and require specialized technical skills for operation and maintenance.

Reclaimed water systems have the highest associated cost due to complex treatment process requirements that ensure recycled water meets regulatory standards for non-potable use and may be difficult for some communities to adopt. Therefore, funding support or subsidies from

USEPA, USDA, U.S Department of Housing and Urban Development (USHUD), U.S. Department of the Interior (USDOl), and FEMA may help with overcoming the financial barrier (USEPA, 2022b). Extensive treatment can address water quality issues; safety concerns may lead to public hesitancy towards acceptance of recycled water use. Both potable and non-potable water reclamation offer promising ways to mitigate disasters such as droughts by expanding local water supplies and is encouraged in many arid states such as California, Arizona, Texas, and Nevada (Ormerod et al., 2020). Conversely, increased dependency on closed-loop process may concentrate pharmaceuticals and other wastewater contaminants that are not easily treated/removed.

The frequency and duration of emergency events can have a significant impact on the ranking of countermeasures and the overall cost-effectiveness. For example, if the frequency and duration of emergency events increase, response cost associated with implementing bulk water and pre-packaged water as the primary response mechanism may become significantly higher. Online water quality monitoring may become more important in the face of more frequent contamination events. Considering the increasing frequency and severity of natural and man-made disasters, there may be long-term cost benefits associated with countermeasures such as local water supply alternatives and interconnections that have higher upfront costs, these measures may be more cost-effective in the long run.

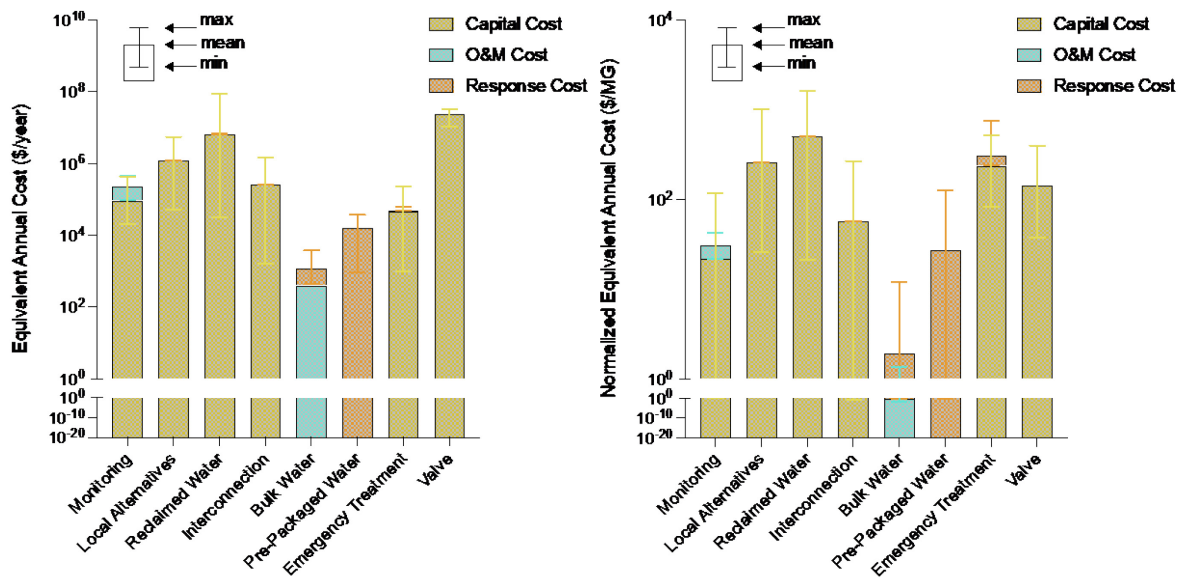


Figure 1. The equivalent annual cost (left) and the normalized equivalent annual cost (right) for each of the countermeasures. Costs consist of 3 components, including capital cost (yellow bar), O&M cost (blue bar), and response cost (orange bar). Each component's contribution to equivalent annual cost is presented in the form of a box plot which displays the distribution of the data for mean, minimum, and maximum.

5. Conclusions

In this study, eight countermeasures for drinking water supply systems, including online water quality monitoring, local water source alternatives, reclaimed water, interconnections, bulk water, pre-packaged water, emergency water treatment, and isolation valves are evaluated from resilience and life cycle cost perspectives. Online water quality monitoring systems offer real-

time data that may identify potential contamination events for rapid response. Although these systems do not require intensive construction investment, trained personnel are needed to interpret data and take appropriate action. However, monitoring alone cannot solve water supply issues and should be combined with other countermeasures to create a more robust and reliable system.

Local alternative water sources can provide backup water during emergencies and promote water supply self-sufficiency with long-term benefits. However, effectiveness of alternate water supply sources can be lower in areas with limited water resources and installation cost can be high.

Reclaimed water is beneficial in water-scarce areas but has high implementation and treatment costs that may pose a financial burden and public acceptance may be challenging.

Interconnection with neighboring water providers is found to be typically less expensive than developing local alternative water sources, but regulatory barriers, control, and management require careful planning. Interconnections alone may not be sufficient to address all emergency situations.

Bulk and pre-packaged water can be used temporarily but are not sustainable, long-term solutions. Emergency water treatment enhances drinking water resilience but requires specialized technical knowledge for operation and maintenance. Water isolation valves enable rapid isolation of potentially contaminated water during emergencies to reduce impacts. However, trained personnel are necessary and complete system shutdown may be necessary without rapid detection and response. While the cost of installation is high, isolation valve installation is cost and time effective for future repairs with less supply disruption to customers.

Overall, countermeasures such as interconnections are considered effective ways to increase resilience from a life cycle cost perspective. There is a clear tradeoff between resilience and life cycle cost. Local water supply alternatives and emergency water treatment are effective countermeasures for quick response and can provide sustained water supply in emergencies. However, these measures may incur higher life cycle costs. Bulk water and pre-packaged water can be limited in supply capacity and may require longer response time; however, life cycle costs were found to be lower for short periods of time. As future drinking water threats are likely to become more frequent and prolonged; it is prudent for water suppliers to invest in countermeasures that perform well in resilience, sustainability, and cost (e.g., interconnections) in addition to planning for the use of integrated countermeasures that combine high capital projects with bulk/pre-packaged water contracts to achieve optimal outcomes. Future studies associated with water system size, resource availability, and specific emergency event scenarios may better support decision-making for countermeasures based on tailored solutions for local water suppliers.

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